SMART INDIA HACKATHON 2025

SAFEROUTETECH

- **Problem Statement ID – 25008**
- **Problem Statement Title- Disaster Preparedness and Response Education System for Schools and Colleges**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal): SafeRouteTech**

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❖ Proposed Solution

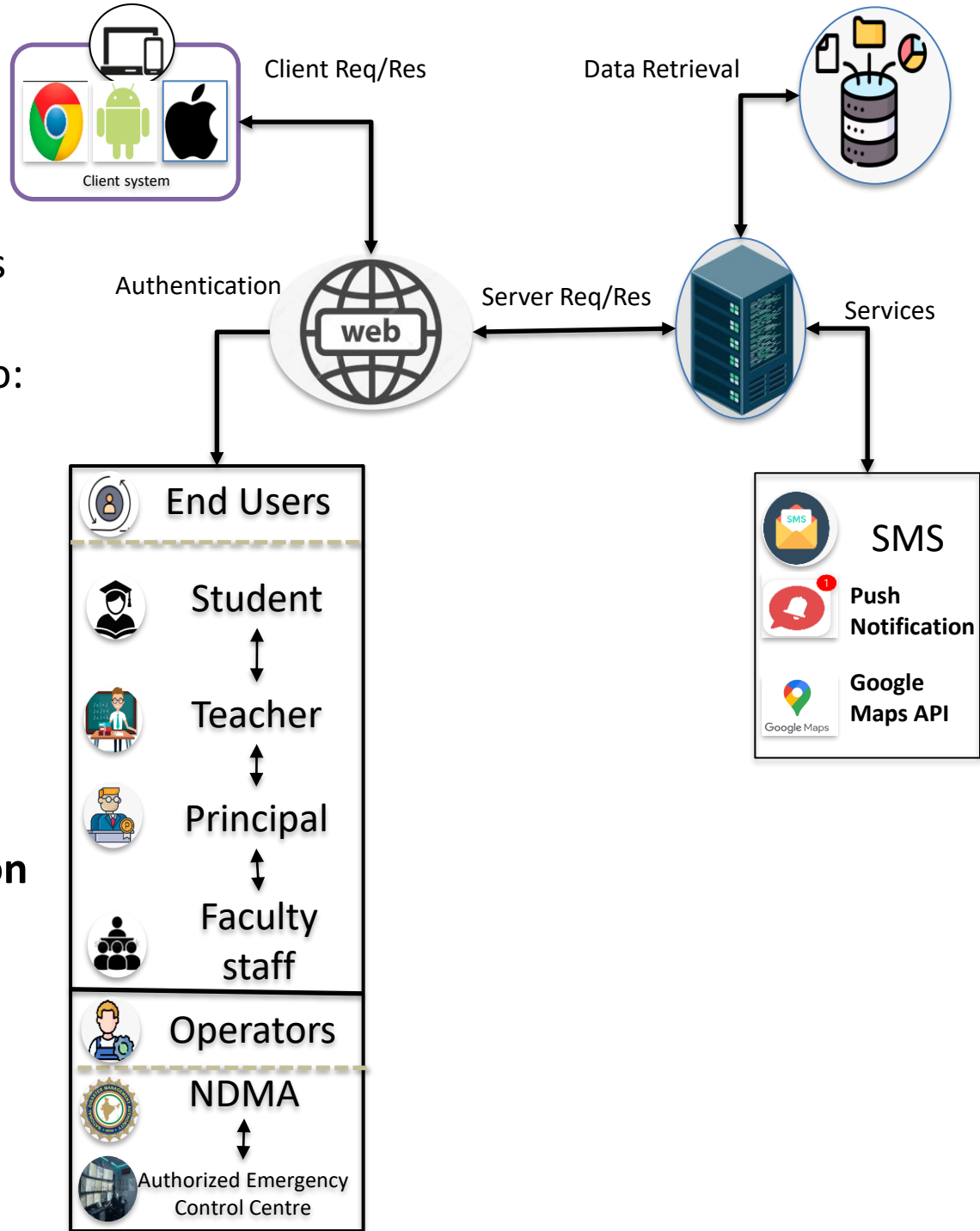
- **Centralized Alert System:** The system continuously monitors **official disaster alerts** (earthquake, flood, cyclone, fire). When a disaster is detected, **instant notifications** are sent to: School and college principals, Teachers and staff.

- **Real-Time Notification:** Notifications are delivered through: **Mobile app**, **SMS** , **Web dashboard.**

- **Evacuation Map:** Directions are provided using **GPS and maps**, making evacuation faster and safer.

- **Training Module:** The system includes **pre-disaster education content** are Disaster safety guidelines, Do's and Don'ts and Mock drill instructions

- **Scalable & Low-Cost Solution:** Can be deployed in Schools, Colleges and Universities. Designed to work even in **low-resource environments**.



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TECHNICAL APPROACH

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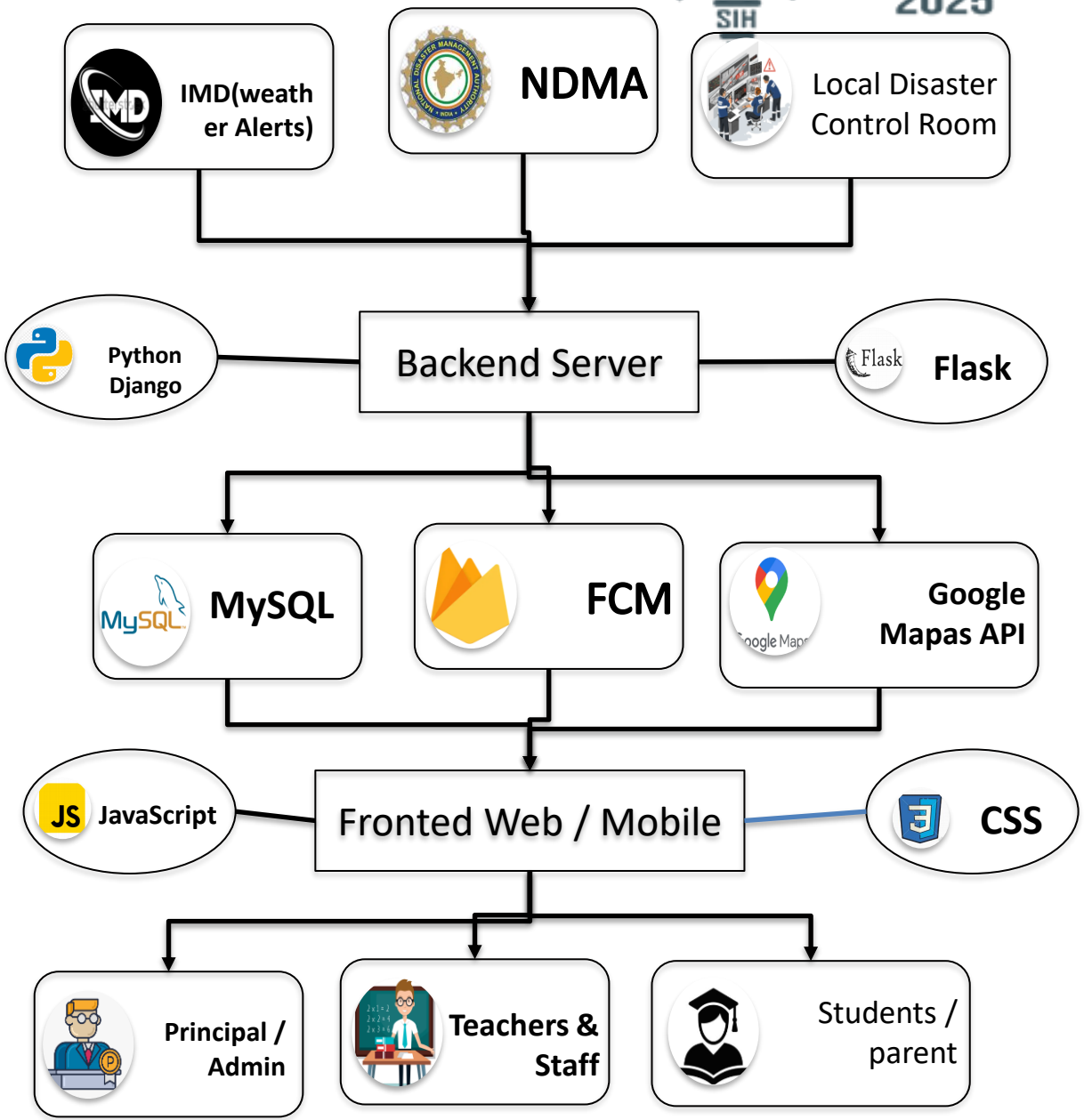
❖ Technology Stack

- **Frontend**: HTML5, CSS3, JavaScript, React.js, Android.
- **Backend**: Python (Django / Flask), REST APIs, Node.js, MySQL, MongoDB, FCM(Firebase Cloud Messaging), Twilio SMS API, Email SMTP Service.
- **API Services**: Google Maps API, OpenStreetMap, GPS Services.
- **Authentication**: JWT(JSON Web Token), RBAC, HTTPS/SSL.
- **Cloud and Deployment**:: AWS/ Google Cloud Platform(GCP), Firebase Hosting, Docker
- **Add Ons**: Prometheus, Grafana, Logstash.



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FEASIBILITY AND VIABILITY

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Feasibility

- Uses existing technologies (mobile apps, cloud, GPS, maps)
- Disaster alerts can be integrated from government or authorized sources
- Real-time notifications via SMS and push alerts are technically reliable
- Easy to deploy and maintain with minimal infrastructure

Viability

- High relevance due to increasing natural disasters
- Suitable for schools and colleges as part of safety compliance
- Scalable from institution level to city or state level
- Long-term usability through regular safety training and updates

Challenges

- Dependence on accurate and timely disaster data
- Internet or power failure during severe disasters
- User training and awareness adoption
- Data privacy and security concerns

Use Cases

- Flood evacuation guidance for schools in low-lying areas
- Earthquake response and safe assembly mapping
- Fire emergency evacuation in campuses
- Mock drills and disaster preparedness training

Business Potential

- Subscription model for schools and colleges
- Government and education board partnerships
- Licensing to private institutions and universities
- Potential integration with insurance and disaster management agencies

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IMPACT AND BENEFITS

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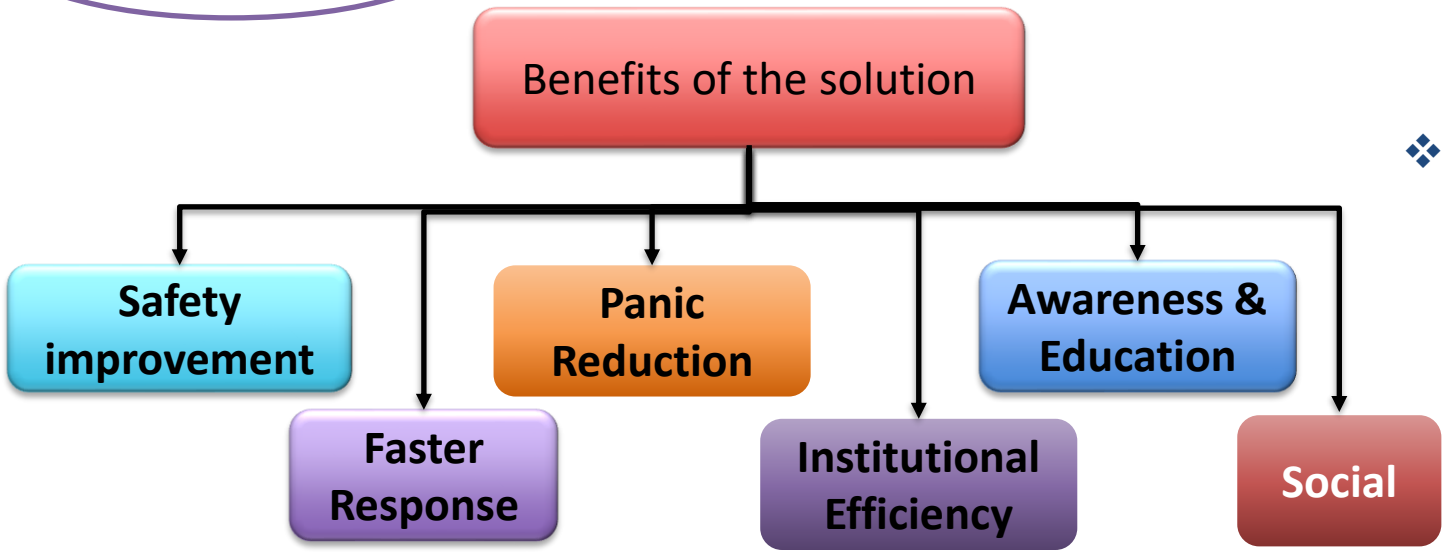

❖ Benefits of the solution

- Provides **real-time disaster alerts** to schools and colleges
- Guides users with **safe evacuation routes using maps**
- Reduces **panic, confusion, and response time** during emergencies
- Improves **student and staff safety**
- Helps teachers and principals take **quick and informed decisions**
- Supports **organized evacuation and crowd management**
- Enhances **disaster awareness and preparedness education**
- Low-cost, scalable, and easy to implement
- Works for **multiple disaster types** (flood, fire, earthquake)
- Builds **trust among parents and authorities**

❖ Potential impact on the target audience

Students

- Faster access to safe evacuation routes
- Reduced panic during emergency situations
- Improved awareness and understanding of disaster safety

Teachers

- Instant disaster notifications and clear instructions
- Better ability to manage and guide students safely

School & College Administration (Principals)

- Centralized control and real-time situation awareness
- Improved compliance with safety regulations
- Faster coordination with local authorities

Parents & Guardians

- Greater trust in institutional safety systems
- Better information flow from institutions

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RESEARCH AND REFERENCES

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Government & Official Sources

National Disaster Management Authority (NDMA) : https://ndma.gov.in
India Meteorological Department (IMD) : https://mausam.imd.gov.in
National Institute of Disaster Management (NIDM) : https://nidm.gov.in

International Organizations

United Nations Office for Disaster Risk Reduction (UNDRR) : https://www.undrr.org
World Health Organization (WHO) : https://www.who.int/emergencies

Technology References

Google Maps Platform Documentation : https://developers.google.com/maps/documentation
Firebase Cloud Messaging (Push Notifications) : https://firebase.google.com/docs/cloud-messaging
OpenStreetMap (Alternative to Google Maps) : https://www.openstreetmap.org

IMPORTANT INSTRUCTIONS

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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
6. **You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.**

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.